

Linear systems and elimination

Monotonic optimal backward error along LSMR

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved

Independently reviewed resolution - 2026-09-11

Negative resolution. Sections 1-4 give one exact full-column-rank 4×3 LSMR example for which both displayed errors increase from the first to the second nonzero iterate. The matrix-only spectral backward error satisfies $\mu(x_1)^2 \leq 1979/2000 < 99/100 < \mu(x_2)^2$, and the specified approximation also strictly increases. The right-hand side stays fixed. This settles the canonical spectral-norm formulation; the cited SISC paper uses a different default norm convention, so no Frobenius-error conclusion is inferred.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects monotonicity of an optimization-defined error along coupled Krylov iterates; community impact is a stopping and reliability guarantee for a widely used least-squares method.

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. In exact arithmetic, start LSMR at $x_0 = 0$: equivalently, x_k minimizes $\|A^T(b - Ax)\|_2$ over $\mathcal{K}_k(A^T A, A^T b)$. Work up to its exact termination and use its minimum-length iterate if necessary. Let $r = b - Ax$ and define the matrix-only normwise backward error

$$\mu(x) = \min\{\|E\|_2 : (A + E)^T((A + E)x - b) = 0\}.$$

For $x \neq 0$, put $K_x = [A^T, (\|r\|_2/\|x\|_2)I]^T$, $v_x = [r^T, 0^T]^T$, and

$$\tilde{\mu}(x) = \frac{\|K_x K_x^\dagger v_x\|_2}{\|x\|_2}.$$

Here $\dagger$ is the Moore–Penrose inverse; at an exact least-squares solution set both errors to zero.

Are both sequences $\mu(x_k)$ and $\tilde{\mu}(x_k)$ nonincreasing, for successive nonzero LSMR iterates? These two closely related claims are counted together, as in the source. The right-hand side b is kept fixed in the backward-error model.

A positive answer would justify backward-error stopping decisions without a later iteration making the current iterate less backward accurate. The known monotonicity of $\|r_k\|_2$ and $\|A^T r_k\|_2$ does not establish this statement.

References

D. C.-L. Fong, *Minimum-Residual Methods for Sparse Least-Squares Using Golub–Kahan Bidiagonalization*, Stanford dissertation 2011, §7.2.1 and definitions (4.1),(4.5), printed pp. 56–57,118 ([primary PDF](#)). Fong and M. Saunders, *LSMR: An iterative algorithm for sparse least-squares problems*, SISC 33(2011),2950–2971, §§6.1–6.2, Fig7.5 ([author PDF](#)). E. Hallman and M. Gu, *LSMB: Minimizing the backward error for least-squares problems*, SIMAX 39(2018),1295–1317 ([author PDF](#)).

Status check — 2026-09-08

Exact-title and “LSMR optimal backward error monotonicity conjecture/proof/counterexample” searches found no resolution. The LSMB2018 discussion of nonmonotonicity concerns LSQR, and its proposed method differs from LSMR. Hallman’s May 2026 arXiv:2605.09211 develops backward-error formulas and estimates, without resolving this iterate-monotonicity conjecture. No recent explicit reaffirmation was located.

Audit update — 2026-09-10

Rechecked Fong’s [dissertation](#), §7.2.1, printed p. 118, which separately conjectures monotonicity of both displayed errors. [Hallman \(2026\)](#) studies a variational equation and lower bound for the backward error; it does not assert this iterate monotonicity. Searches for LSMR optimal-backward-error monotonicity located no proof or counterexample.